

Ionospheric electron concentration imaging using GPS over the USA during the storm of July 2000

P. Yin,¹ C. N. Mitchell,¹ P. S. J. Spencer,¹ and J. C. Foster²

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[1] A new technique for ionospheric imaging is demonstrated during a severe geomagnetic storm of 15th July 2000. The three-dimensional time-dependent imaging algorithm is applied to multi-directional ground-based GPS data and yields spatial maps of electron concentration. This technique is demonstrated by showing a series of images of the mid-latitude ionosphere over the USA during the storm of July 2000. A strong uplift in the height of the F-layer is observed after 20 UT on 15th July, followed by a severe latitudinal electron concentration gradient. Independent verification of the images is provided by the ionosondes and incoherent scatter radar data. The GPS images reveal the large-scale dynamics of the ionosphere during the disturbed conditions and show the potential of geophysical imaging for storm-time ionospheric studies. **INDEX TERMS:** 2435 Ionosphere: Ionospheric disturbances; 2437 Ionosphere: Ionospheric dynamics; 2494 Ionosphere: Instruments and techniques. **Citation:** Yin, P., C. N. Mitchell, P. S. J. Spencer, and J. C. Foster (2004), Ionospheric electron concentration imaging using GPS over the USA during the storm of July 2000, *Geophys. Res. Lett.*, 31, L12806, doi:10.1029/2004GL019899.

1. Introduction

[2] Severe space-weather events impact on the interplanetary- and geo-magnetic fields, electric fields and neutral winds. The response of the ionosphere is characterised by rapid temporal changes and steep spatial gradients in the electron concentration. During the most disturbed conditions severe storms have a global impact on the ionisation, with dramatic changes in latitudinal morphology and abrupt changes in layer height occurring. While the main phase of a storm may last less than a day the entire recovery time, accounting for the dynamical, thermal and compositional changes, can last for many days. A review of ionospheric storms is given by *Buonsanto* [1999]. Due to the global extent of the events and the strong gradients involved it is advantageous to study these physical processes with measurements that are near continuous in latitude, longitude, altitude and time. The possibility to realize this goal has recently developed out of tomography and the Global Positioning System (GPS).

[3] Two-dimensional tomography, using Low-Earth-Orbit (LEO) satellite data, is an established technique to image the electron concentration [*Bernhardt et al.*, 1998].

Conventionally, ionospheric tomography uses differential Doppler measurements recorded at a special longitudinal chain of receivers from satellites in polar low-earth-orbit. The network of total electron content (TEC) measurements is then inverted into images of electron concentration. The new TEC data source, from dual-frequency GPS receivers, has the advantage that arrays of receivers are already operated across the world for geodetic measurements and atmospheric science. These GPS-to-ground signals trace out a network of measurements through the ionosphere. However, the geometry does not map easily into a plane for tomographic inversion and the need to extend conventional imaging techniques to three- and four-dimensional imaging has arisen [*Bust et al.*, 2001; *Hernandez-Pajares et al.*, 2000]. New inversion algorithms, such as Multi-Instrument Data Analysis System (MIDAS) [*Mitchell and Spencer*, 2003], can create time-dependent images of the electron concentration from inhomogeneous and diverse instrumentation. Here this geophysical imaging technique is applied to GPS data to image the distribution of electron concentration over the USA during the storm of July 2000.

2. Method

[4] GPS observations from a world - wide network of receivers have been made available to the scientific community by the international GPS service (IGS). Figure 1 shows the locations of the GPS receivers that were used for this study of the ionosphere over the USA. The GPS data used in this study have been analysed using the MIDAS algorithms, which are well described in *Mitchell and Spencer* [2003] and only a brief description is given here. In summary, dual-frequency radio signals that propagate through the ionosphere are subject to a differential phase change due to the dispersive nature of the plasma. As a first-order approximation the change in the differential phase shift is directly proportional to the change in TEC between the transmitter and receiver. The method uses the differential-phase observations to create a matrix of relative slant-TEC observations. Provided that there is a continuous record of observations relative TEC can be derived from differential-phase for a particular satellite-to-receiver pair. In general, these relative TECs are used as the input data to the inversion algorithm.

[5] Under quiet/moderate conditions with good data coverage and few breaks in the phase records the calibration of the absolute TEC is well determined within the inversion. However, under very disturbed conditions, such as were experienced during 15th July 2000, it is advantageous to pre-calibrate the TEC. Hence for analysis of GPS data leading to storm images a different procedure is adopted. Firstly, the satellite-receiver inter-frequency biases were

¹Department of Electronic and Electrical Engineering, University of Bath, Bath, UK.

²Haystack Observatory, Massachusetts Institute of Technology, Westford, Massachusetts, USA.

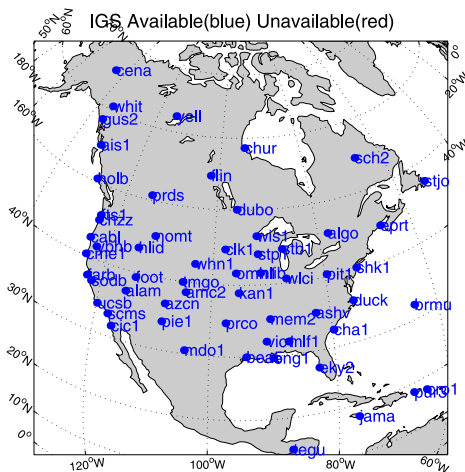


Figure 1. The locations of the GPS receivers used in the imaging.

determined from a preliminary set of MIDAS inversions using GPS raw observations for the storm days. The mean values of the inter-frequency biases over three storm days (14th, 15th, 16th July 2000) were found. These inter-frequency biases were subsequently used to calibrate each of the differential-delay observations for each satellite-receiver pair. Then a second set of MIDAS inversions were computed using the bias-corrected differential-delay observations in a least-square fitting to calibrate the differential-phases. Thus the absolute slant TECs during the 15th July storm were determined from the differential phase and time

delays recorded during the storm and the inter-frequency biases from before.

[6] The data were sampled at 30 seconds, then partitioned into one-hour sets and three-dimensional inversions with a linear time evolution were performed, resulting in a series of hour-long movies of the electron concentration. The resulting images extended latitudinally from 30°N to 90°N and longitudinally from 140°W to 60°W. The altitudes ranged from 80 km to 1200 km. In addition to the GPS data there were also other ground-based instruments active during the storm. The Millstone Hill Incoherent Scatter Radar (ISR) was being operated in a wide-coverage scanning mode throughout the July 2000 storm event. South to north elevation scans with the steerable MISA antenna were run with a ~60-min repetition interval. Each scan provided direct observations of ionospheric density, ion and electron temperatures, and line of sight ion velocity. The scan plane lay along 78°W longitude and F-region parameters were obtained up to 1000 km altitude and at latitudes extending slightly below 25°S geodetic. At the furthest reach to the south, the Millstone Hill scans sampled the edge of the greatly-expanded equatorial anomaly region during this event, while a deep ionospheric trough formed at latitudes poleward of 30°N. These ISR values were compared to evaluate the GPS results. A number of ionosondes also operating from mainland USA and NmF2 data are shown for comparisons.

3. Results

[7] Figures 2a–2d shows the equivalent vertical TECs for the last four hours of 15th July. These TECs were

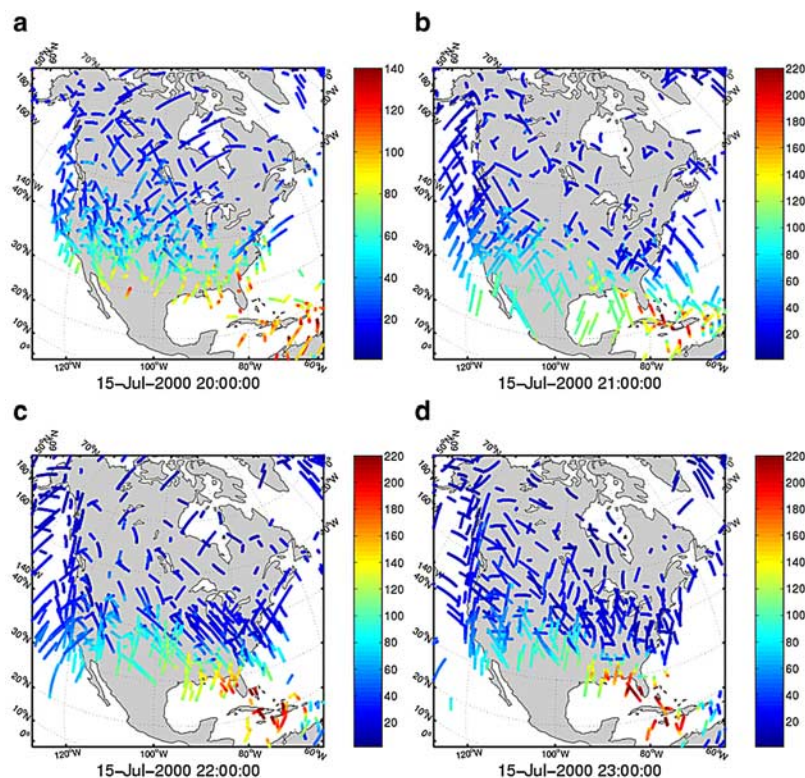


Figure 2. Equivalent vertical TECs for (a) 20–21 UT, (b) 21–22 UT, (c) 22–23 UT and (d) 23–24 UT on 15th July 2000. (Note: TECU is up to 140 at 20–21 UT, but 220 during other times).

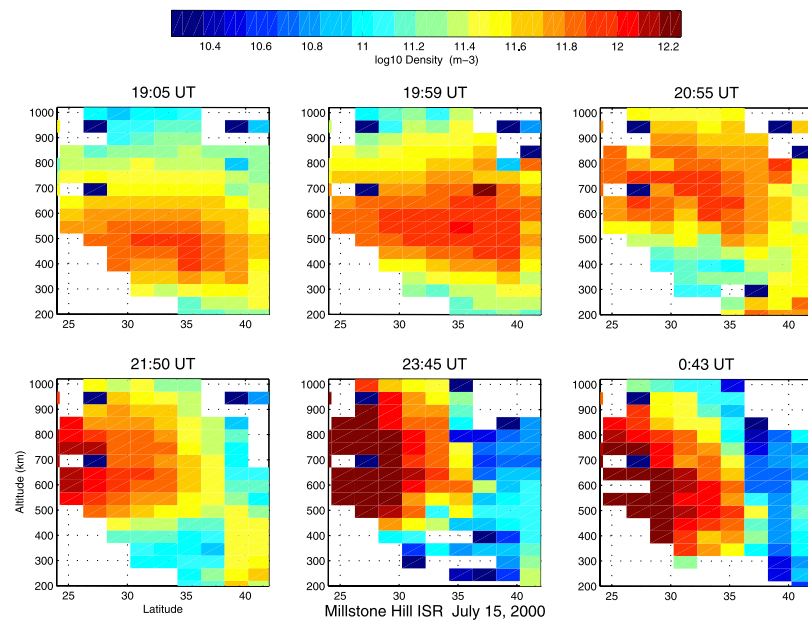


Figure 3. Millstone Hill ISR elevation scans on 15th July 2000.

obtained by assuming a pierce point with a shell at 400 km altitude and making a geometrical correction from the bias-corrected slant TEC. Although the MIDAS algorithms do not require a shell approximation it is useful to show the vertical TEC here in this manner to show the distribution of observations collected during each one-hour period. In addition, it is interesting to see that there is a strong enhancement in TEC beyond the coast of the southeast USA. The values of equivalent vertical TEC that we have calculated from ray paths extending off the southeast coast were well in excess of 220 TECU ($1 \text{ TECU} = 10^{16} \text{ m}^{-2}$). During these very disturbed conditions the geomagnetic index Kp reached 9.

[8] Figure 3 shows latitude-altitude maps of electron concentration observed with the Millstone Hill ISR which depict the evolution of the mid/low latitude ionosphere

during the 15th July 2000 event. Each scan to the south of Millstone Hill was obtained during a 10-min interval and scans were repeated at a 1-hour repetition interval. It is clearly seen that F-region peak altitude is considerably uplifted to ~ 700 km near 21 UT. After this uplift event, the peak stays high until 00 UT and the density and TEC in this low-mid latitude region becomes very large. The radar scans also observe the downward field-aligned collapse of the greatly-enhanced ionization seen at and equatorward of 30°N at and after 21 UT. This event coincides with the steep wall of ionisation seen in the cross-sectional images of Figure 4. The series of GPS images are taken from the first frame of each one-hour movie, at a longitude of 72°W where the Millstone Hill ISR is located. Increasing electron concentrations can be seen in the four images between 20 UT on 15th to 00 UT on 16th. Around this time the

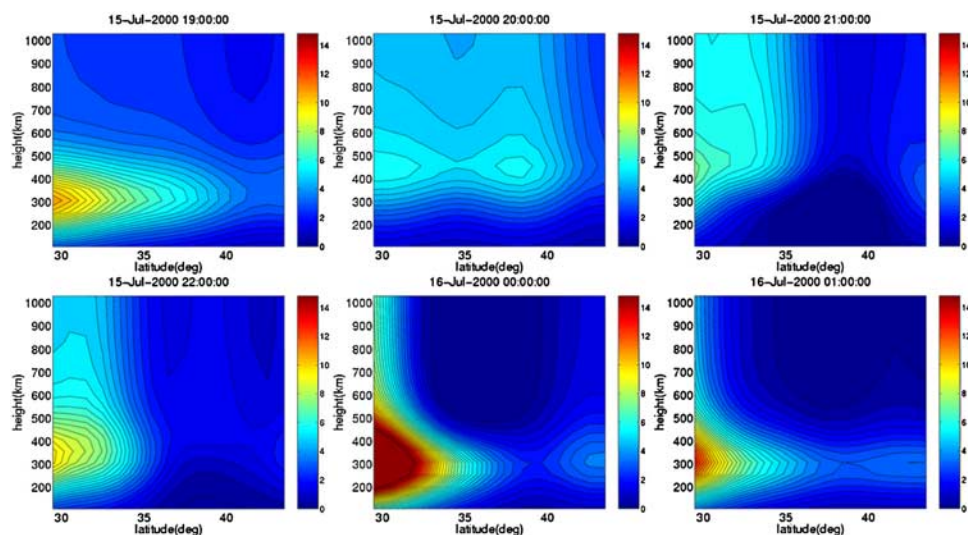


Figure 4. Variation in electron concentration ($\times 10^{11} \text{ m}^{-3}$) at 72°W (Millstone Hill) for the 19 UT (15th)–01 UT (16th) period.

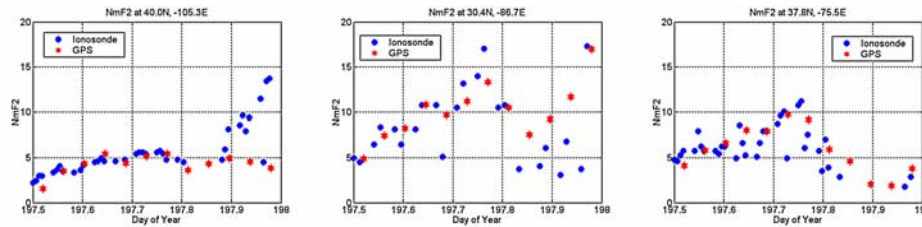


Figure 5. Comparison of $NmF2$ ($\times 10^{11} \text{ m}^{-3}$) from ionosondes (points) and inversions (stars) using ground-based GPS data for the 12–24 UT hours of the day 197 (15th July 2000).

positive storm effects evolved into extreme steep electron-density and TEC gradients to the south of Florida. This event was associated with the upwelling of the entire ionization, evidenced by hugely enhanced peak heights and topside content. The reason is probably that a strong penetrating eastward electric field (near 21 UT) uplifts the low latitude ionosphere resulting in a serious equatorial depletion [Basu *et al.*, 2001] and a poleward displacement of an enhanced equatorial anomaly to latitudes near Florida, as discussed for previous storm events by [Foster and Rich, 1998]. It is interesting to note that the GPS reconstructed images indicate that the peak height has lowered at around 22 UT. It is possible that this is a loss of height resolution in our images as the uplift event nears the edge of our data coverage and imaging region. Figure 5 shows the peak electron concentrations ($NmF2$) derived from the ionosonde foF2s and the MIDAS images at the locations of the ionosondes. It can be seen that there is a general agreement with an obvious divergence at 21 UT on the three locations due to the steep wall of ionisation. Indeed the ionosondes are almost certainly receiving echoes from the off-vertical path to the south of the sounders around this time.

4. Summary

[9] Two-dimensional tomographic methods originally applied to the reconstruction of transit satellite observations have been extended to three-dimensional time-dependent imaging and applied to ground based GPS observations. GPS observations are generally continuous in space and time, enabling the evolution of large-scale ionospheric features to be studied. The time-dependent inversion applied here is important because of temporal changes in the ionosphere during the data collection period.

[10] The imaging technique has been demonstrated for a highly disturbed storm period during 15th July 2000. The images showed very steep gradients in electron concentration and TEC over the USA. Extreme peak heights and density gradients in the images were confirmed by the Millstone Hill ISR data. Comparisons with ionosondes peak densities diverged during the evening of 15th July, when extreme TEC values and gradients were observed. This event was associated with a sudden upwelling of the ionization. The results demonstrate the potential of a new

analysis technique allowing GPS data to be used for large-scale studies of the ionosphere under very disturbed geomagnetic conditions.

[11] A loss of height resolution was experienced at the edge of the imaging region. Hajj *et al.* [1994] have suggested the incorporation of GPS-LEO radio occultation data into ionospheric images to improve the vertical resolution and geographical coverage of measurements. Although there were no GPS-LEO available for this particular period there are now a number of such operational LEOs and it is expected that we can test the use of such data for improving the coverage and vertical resolution for imaging during other storms from the current solar maximum.

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C. N. Mitchell, P. S. J. Spencer, and P. Yin, Department of Electronic and Electrical Engineering, University of Bath, Bath BA2 7AY, UK. (p.yin@bath.ac.uk)

J. C. Foster, Haystack Observatory, Massachusetts Institute of Technology, Westford, MA, USA.